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Neural Networks for Quantum-System Simulation

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Lecture notes

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Chapter 1

Variational Monte Carlo

The simulation of interacting quantum many-body systems remains a central challenge of computational physics. The difficulty originates from the exponential growth of the Hilbert space with system size. A generic pure state can be written as¹

$$|\psi\rangle = \sum_x \psi(x) |x\rangle, \quad (1.1)$$

where the number of amplitudes $\psi(x)$ grows exponentially with the number of degrees of freedom, making exact representations rapidly intractable.

Several complementary approaches have been developed to address this challenge. Exact diagonalisation is unbiased but limited to small Hilbert spaces. Quantum Monte Carlo methods can access very large system sizes, but their applicability is often limited by the sign problem encountered in frustrated spin and fermionic systems [1]. Tensor-network methods exploit the entanglement structure of quantum states and have achieved remarkable success across a wide range of quantum many-body problems [2, 3]. Their computational cost, however, generally increases rapidly with the amount of entanglement that must be represented [4–7], making simulations of highly entangled states, long-time dynamics, or higher-dimensional systems generally challenging [8–11].

A broadly applicable alternative is provided by variational methods, which recast the ground-state problem as the optimisation of a parametrised wavefunction. Since exact evaluation of the resulting objective remains prohibitively expensive, variational Monte Carlo (VMC) estimates expectation values through statistical sampling of the wavefunction-derived probability distribution [12–14]. The success of this approach ultimately depends on the expressive power of the chosen variational family.

Since their introduction by Carleo and Troyer [15], neural quantum states (NQS) have emerged as one of the most successful variational frameworks for quantum many-body simulation. The central idea is to represent the many-body wavefunction using a neural network whose parameters are optimised within the VMC framework. Originally based on restricted Boltzmann machines, NQS now encompass a broad range of modern machine-learning architectures and have achieved state-of-the-art accuracy in frustrated spin systems [16–18]. Beyond spin models, they have been successfully applied to all sorts of quantum degrees of freedom,

¹Throughout this chapter, we will use discrete notations for simplicity.

including bosons [19–22], fermions [23–25], and beyond [26, 27]. Their scope now extends beyond ground-state calculations to excitation spectra [19, 28, 29] and quantum dynamics [30–34].

From a theoretical perspective, NQS are particularly attractive because, unlike tensor-networks, their expressive power is not known to be fundamentally limited by the amount of entanglement present in the target state. Both analytical [35–40] and numerical studies [35, 40, 41] have demonstrated their ability to efficiently represent highly entangled quantum states. Neural quantum states therefore combine the expressive power of modern neural networks with the generality of the VMC framework, providing a powerful and broadly applicable approach to quantum many-body simulation.

1.1 Variational approach to the quantum many-body problem

One of the central objectives of quantum many-body physics is to solve the Hamiltonian eigenvalue problem

$$\hat{H}|\psi_k\rangle = E_k|\psi_k\rangle. \quad (1.2)$$

In particular, throughout these lectures, we will focus on the determination of the ground state of a Hamiltonian $\hat{H}$. Formally, this corresponds to finding the lowest-energy eigenpair satisfying

$$\hat{H}|\psi_0\rangle = E_0|\psi_0\rangle, \quad (1.3)$$

where E_0 denotes the ground-state energy.

1.1.1 The variational principle

Consider an arbitrary normalisable quantum state $|\psi\rangle$. Its energy expectation value is given by the Rayleigh quotient

$$E[\psi] = \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle}. \quad (1.4)$$

Expanding the state in the eigenbasis of the Hamiltonian,

$$|\psi\rangle = \sum_n c_n |\psi_n\rangle, \quad (1.5)$$

one obtains

$$E[\psi] = \frac{\sum_n |c_n|^2 E_n}{\sum_n |c_n|^2}. \quad (1.6)$$

Since all eigenvalues satisfy $E_n \geq E_0$, it immediately follows that

$$E[\psi] \geq E_0. \quad (1.7)$$

This statement is known as the variational principle. It implies that the ground-state energy can be obtained as the global minimum of the energy functional:

$$E_0 = \min_{|\psi\rangle} E[\psi]. \quad (1.8)$$

Unlike many optimisation problems encountered in machine learning, this objective possesses a physically meaningful global optimum corresponding to the exact ground state.

1.1.2 Variational manifold

The optimisation problem of Eq. (1.8) is formally defined over the entire Hilbert space. For generic interacting systems, however, the number of degrees of freedom required to specify an arbitrary quantum state grows exponentially with system size, making the exact optimisation problem intractable. In continuous-space systems, an exact representation furthermore requires an infinite-dimensional basis.

The central idea of variational methods is that physically relevant states occupy a highly structured subset of Hilbert space. The exploitation of this structure underlies the success of modern wavefunction Ansätze. For instance, tensor-network states leverage the entanglement structure of ground states of local Hamiltonians to obtain efficient parametrisations using only polynomially many variational parameters.

The exact quantum state is replaced by a parametrised variational wavefunction $|\psi_{\boldsymbol{\theta}}\rangle$, where the vector $\boldsymbol{\theta}$ contains the variational parameters to be optimised. More generally, one restricts the optimisation to a variational manifold

$$\mathcal{M} = \{|\psi_{\boldsymbol{\theta}}\rangle \mid \boldsymbol{\theta} \in \mathbb{R}^p\}, \quad (1.9)$$

with p scaling polynomially with system size, and seeks

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} E_{\boldsymbol{\theta}}, \quad (1.10)$$

where we introduced the important quantity of *variational energy*:

$$E_{\boldsymbol{\theta}} = \frac{\langle \psi_{\boldsymbol{\theta}} | \hat{H} | \psi_{\boldsymbol{\theta}} \rangle}{\langle \psi_{\boldsymbol{\theta}} | \psi_{\boldsymbol{\theta}} \rangle}. \quad (1.11)$$

The accuracy of the resulting approximation is determined by the expressive power of the chosen variational manifold.

1.11 Monte Carlo estimation of the variational energy

An important point before discussing any strategy to minimise the variational energy is to be able to estimate the variational energy $E_{\boldsymbol{\theta}}$ with a complexity scaling polynomially with system size.

Upon inserting the resolution of the identity in the computational basis, we can express this quantity as

$$E_{\boldsymbol{\theta}} = \frac{\sum_{x,y} \psi_{\boldsymbol{\theta}}^*(x) H_{xy} \psi_{\boldsymbol{\theta}}(y)}{\sum_x |\psi_{\boldsymbol{\theta}}(x)|^2}, \quad (1.12)$$

where

$$H_{xy} := \langle x | \hat{H} | y \rangle. \quad (1.13)$$

At this stage, the direct evaluation of this expression requires summations over the full Hilbert space and is therefore generally intractable.

Let us introduce the variational Born probability over configurations.

$$p_{\boldsymbol{\theta}}(x) := \frac{|\psi_{\boldsymbol{\theta}}(x)|^2}{\sum_{x'} |\psi_{\boldsymbol{\theta}}(x')|^2} \quad (1.14)$$

and further define the so-called *local energy*, i.e. the local estimator of the variational energy, as

$$E_{\text{loc}}(x) := \frac{\langle x | \hat{H} | \psi_{\theta} \rangle}{\langle x | \psi_{\theta} \rangle} \equiv \sum_y H_{xy} \frac{\psi_{\theta}(y)}{\psi_{\theta}(x)}. \quad (1.15)$$

Using these definitions, one finally obtains

$$E_{\theta} = \sum_x p_{\theta}(x) E_{\text{loc}}(x), \quad (1.16)$$

that is, the variational energy can be obtained as a *classical* expectation value of the local energy.

Two operations are still not transparently of polynomial complexity: the evaluation of the local energy and the sum over all configurations.

A first important simplification arises upon exploiting the sparsity of the Hamiltonian and writing

$$E_{\text{loc}}(x) = \sum_{y \in \mathcal{C}_H(x)} H_{xy} \frac{\psi_{\theta}(y)}{\psi_{\theta}(x)}, \quad (1.17)$$

where $\mathcal{C}_H(x) = \{y | \langle x | \hat{H} | y \rangle \neq 0\}$ denotes the *connected elements* of the Hamiltonian with respect to configuration x . For local Hamiltonians, the evaluation of E_{loc} is thus of constant complexity.

A second more profound simplification consists in replacing the full summation over configurations weighted by their corresponding variational Born weights by a Monte Carlo integration:

$$\hat{E}_{\theta} = \frac{1}{N_s} \sum_{i=1}^{N_s} E_{\text{loc}}(x^{(i)}), \quad (1.18)$$

where the samples $\{x^{(i)}\}_{i=1}^{N_s}$ are independently drawn from p_{θ} . The Monte Carlo estimate $\hat{E}_{\theta}$ is therefore a random variable whose value depends on the particular realisation of the sampling procedure.

By construction, the estimator is unbiased,

$$\mathbb{E}[\hat{E}_{\theta}] = E_{\theta}, \quad (1.19)$$

where the expectation is taken over all possible sampling outcomes. Furthermore, the law of large numbers guarantees that

$$\hat{E}_{\theta} \xrightarrow[N_s \rightarrow \infty]{} E_{\theta} \quad (1.20)$$

almost surely. Finally, the statistical fluctuations of the estimator are characterised by

$$\text{Var}[\hat{E}_{\theta}] = \frac{\text{Var}_{p_{\theta}}[E_{\text{loc}}]}{N_s}, \quad (1.21)$$

such that the statistical error decreases as

$$\Delta E_{\theta} \sim \frac{\sqrt{\text{Var}_{p_{\theta}}[E_{\text{loc}}]}}{\sqrt{N_s}}. \quad (1.22)$$

1.II.1 The zero-variance principle

The statistical complexity of the Monte Carlo estimator is entirely controlled by the variance of the local energy. Remarkably, this quantity possesses a special property which distinguishes variational Monte Carlo from many other stochastic optimisation methods.

Suppose that the variational state is an exact eigenstate of the Hamiltonian,

$$\hat{H}|\psi_{\theta}\rangle = E|\psi_{\theta}\rangle. \quad (1.23)$$

Then, for every configuration x satisfying $\psi_{\theta}(x) \neq 0$,

$$E_{\text{loc}}(x) = \frac{\langle x | \hat{H} | \psi_{\theta} \rangle}{\langle x | \psi_{\theta} \rangle} = E. \quad (1.24)$$

The local energy therefore becomes independent of the sampled configuration and coincides exactly with the corresponding eigenvalue. As a result,

$$\text{Var}_{p_{\theta}}[E_{\text{loc}}] = 0. \quad (1.25)$$

In particular, if the variational state coincides with the ground state, the Monte Carlo estimator recovers the exact ground-state energy with zero statistical uncertainty. As the variational state approaches an eigenstate, the statistical fluctuations of the energy estimator decrease and Monte Carlo estimates become increasingly accurate. More generally, the variance of the local energy provides a direct measure of the quality of the variational wavefunction.

1.III Monte Carlo sampling

The Monte Carlo estimators introduced above assume the availability of samples drawn from the variational Born distribution

$$p_{\theta}(x) = \frac{|\psi_{\theta}(x)|^2}{\sum_y |\psi_{\theta}(y)|^2}. \quad (1.26)$$

For generic many-body systems, however, the configuration space grows exponentially with system size, making direct sampling generally impossible. We shall therefore rely on Markov-chain Monte Carlo (MCMC) techniques [42].

The basic idea is to construct a Markov chain

$$x^{(0)} \rightarrow x^{(1)} \rightarrow x^{(2)} \rightarrow \dots \quad (1.27)$$

whose stationary distribution coincides with the target distribution $p_{\theta}(x)$. For this to hold, the chain must satisfy three key properties:

(i) **Normalisation:**

$$\sum_y T(x \rightarrow y) = 1, \quad (1.28)$$

where $T(x \rightarrow y)$ denotes the transition probability.

(ii) **Ergodicity:** every configuration with non-zero probability must be reachable from every other configuration through a finite sequence of moves.

(iii) **Stationarity:**

$$\sum_x p_{\theta}(x)T(x \rightarrow y) = p_{\theta}(y). \quad (1.29)$$

A sufficient condition for stationarity is the stronger detailed-balance condition

$$p_{\theta}(x)T(x \rightarrow y) = p_{\theta}(y)T(y \rightarrow x), \quad (1.30)$$

which is satisfied by the Metropolis-Hastings algorithm [43, 44] the most common sampling strategy in variational Monte Carlo.

Starting from a configuration x , a candidate configuration y is proposed according to a proposal distribution $q(x \rightarrow y)$. The move is then accepted with probability

$$A(x \rightarrow y) = \min\left(1, \frac{p_{\theta}(y)q(y \rightarrow x)}{p_{\theta}(x)q(x \rightarrow y)}\right). \quad (1.31)$$

For symmetric proposal distributions satisfying

$$q(x \rightarrow y) = q(y \rightarrow x), \quad (1.32)$$

the acceptance probability reduces to [45]

$$A(x \rightarrow y) = \min\left(1, \frac{|\psi_{\theta}(y)|^2}{|\psi_{\theta}(x)|^2}\right), \quad (1.33)$$

where the normalisation of the wavefunction cancels exactly. This property makes Metropolis sampling particularly attractive in the variational setting.

Because consecutive samples are generated through local updates of a Markov chain, they are generally correlated. As a result, the effective number of independent samples is smaller than the total number of generated configurations. This effect is quantified by the autocorrelation time τ_{corr} , which measures the typical number of Monte Carlo steps required to produce a statistically independent configuration. The variance of Monte Carlo estimators is therefore increased according to

$$\text{Var}[\hat{E}_{\theta}] = \frac{\tau_{\text{corr}}}{N_s} \text{Var}[E_{\text{loc}}]. \quad (1.34)$$

Equivalently, a Markov chain of length N_s contains only approximately N_s/τ_{corr} statistically independent samples.

Efficient sampling strategies therefore seek to minimise both the variance of the local energy and the autocorrelation time of the Markov chain. In practical variational Monte Carlo calculations, sampling is typically performed using many Markov chains in parallel. This improves hardware utilisation and reduces the correlation between successive optimisation steps.

Furthermore, kept configurations are often separated by several Monte Carlo updates, commonly referred to as a sweep. The sweep length is typically chosen as to reduce correlations between successive kept configurations, while residual autocorrelations are accounted for when

estimating statistical uncertainties. Underestimating autocorrelation effects generally leads to overconfident uncertainty estimates and may adversely affect the optimisation procedure.

While Markov-chain Monte Carlo remains the standard sampling strategy for generic neural quantum states, some architectures permit direct sampling from the Born distribution. In particular, autoregressive neural quantum states generate independent samples exactly distributed according to p_{θ} , thereby eliminating autocorrelation effects altogether.

1.IV Optimising the variational energy

Now that we know how to efficiently estimate the variational energy, the variational principle of Eq. (1.10) suggests a direct optimisation strategy analogous to standard machine-learning practice. Given a differentiable variational Ansatz $|\psi_{\theta}\rangle$, one may minimise the variational energy by iteratively following its gradient,

$$\theta_{n+1} = \theta_n - \eta \nabla_{\theta} E_{\theta}, \quad (1.35)$$

where η denotes the learning rate.

This update requires the evaluation of the energy gradient. To derive a practical estimator, it is convenient to introduce the logarithmic derivatives

$$O_k(x) := \frac{\partial_{\theta_k} \psi_{\theta}(x)}{\psi_{\theta}(x)} = \partial_{\theta_k} \log \psi_{\theta}(x). \quad (1.36)$$

Differentiating the variational energy and using the same manipulations as in the derivation of the local-energy estimator yields

$$\partial_k E_{\theta} = \langle O_k^* E_{\text{loc}} \rangle - \langle O_k^* \rangle \langle E_{\text{loc}} \rangle + \text{c.c.}, \quad (1.37)$$

or equivalently

$$\partial_k E_{\theta} = 2\text{Re}[\langle O_k^* E_{\text{loc}} \rangle - \langle O_k^* \rangle \langle E_{\text{loc}} \rangle]. \quad (1.38)$$

The energy gradient is therefore expressed as a covariance between the logarithmic derivatives and the local energy. In particular, it can be estimated using the same Monte Carlo samples employed for the evaluation of the energy itself.

At this stage, the variational ground-state search problem appears formally very similar to a standard machine-learning optimisation problem. The wavefunction defines a parametrised model, the variational energy acts as a loss function, and both the objective and its gradients can be estimated stochastically from samples.

The analogy is, however, incomplete. In contrast to conventional supervised learning, the samples are not drawn from a fixed dataset but from the variational distribution itself. Furthermore, as discussed in the previous section, the statistical fluctuations of the estimators vanish as the variational state approaches an exact eigenstate. Finally, the variational manifold inherits a natural geometry from the Hilbert space in which it is embedded. Ordinary gradient descent ignores this geometric structure and treats parameter space as Euclidean.

The latter observation is particularly important. The Euclidean distance between two parameter vectors does not, in general, provide a meaningful measure of the distance between

the corresponding quantum states. As a result, optimisation algorithms based solely on the parameter-space gradient may follow trajectories that are poorly adapted to the geometry of the variational manifold. This motivates the introduction of geometry-aware optimisation schemes, the most widely used of which is *stochastic reconfiguration* [17, 18, 46–48].

1.IV.1 Stochastic reconfiguration

The discussion above suggests a natural question: can one exploit the geometric structure of the variational manifold to construct optimisation algorithms better adapted to the quantum many-body problem than ordinary gradient descent?

To answer this question, it is useful to return to the original optimisation problem in Hilbert space. Recall that the ground state can be obtained through imaginary-time evolution,

$$\frac{\partial}{\partial \tau} |\psi(\tau)\rangle = -\hat{H} |\psi(\tau)\rangle, \quad (1.39)$$

whose formal solution reads

$$|\psi(\tau)\rangle = e^{-\tau \hat{H}} |\psi(0)\rangle. \quad (1.40)$$

Expanding the initial state in the eigenbasis of the Hamiltonian,

$$|\psi(0)\rangle = \sum_n c_n |\psi_n\rangle, \quad \hat{H} |\psi_n\rangle = E_n |\psi_n\rangle, \quad (1.41)$$

one obtains

$$|\psi(\tau)\rangle = \sum_n c_n e^{-\tau E_n} |\psi_n\rangle. \quad (1.42)$$

After normalisation,

$$\frac{|\psi(\tau)\rangle}{\|\psi(\tau)\|} = \frac{c_0 |\psi_0\rangle + \sum_{n>0} c_n e^{-\tau(E_n - E_0)} |\psi_n\rangle}{\|c_0 |\psi_0\rangle + \sum_{n>0} c_n e^{-\tau(E_n - E_0)} |\psi_n\rangle\|}. \quad (1.43)$$

Provided that $c_0 \neq 0$, the excited-state contributions are exponentially suppressed and the wavefunction converges asymptotically towards the ground state. The convergence rate is controlled by the spectral gap $\Delta = E_1 - E_0$, with characteristic timescale $1/\Delta$.

In practice, however, the wavefunction is constrained to belong to the variational manifold

$$\mathcal{M} = \{|\psi_{\boldsymbol{\theta}}\rangle \mid \boldsymbol{\theta} \in \mathbb{R}^p\}. \quad (1.44)$$

The exact imaginary-time trajectory therefore cannot, in general, be followed exactly.

Consider an infinitesimal parameter update

$$\boldsymbol{\theta} \rightarrow \boldsymbol{\theta} + \delta \boldsymbol{\theta}. \quad (1.45)$$

To first order, the corresponding wavefunction variation is

$$|\psi_{\boldsymbol{\theta} + \delta \boldsymbol{\theta}}\rangle = |\psi_{\boldsymbol{\theta}}\rangle + \sum_k \delta \theta_k \partial_k |\psi_{\boldsymbol{\theta}}\rangle + \mathcal{O}(\delta \theta^2). \quad (1.46)$$

The admissible directions of motion are therefore restricted to the tangent space spanned by the vectors

$$\{\partial_k |\psi_{\boldsymbol{\theta}}\rangle\}_{k=1}^p. \quad (1.47)$$

The central idea of stochastic reconfiguration is to choose the parameter update whose induced variation of the wavefunction best approximates an infinitesimal imaginary-time step within this tangent space.

To leading order in $\delta\tau$, the exact imaginary-time evolution is

$$|\psi'\rangle = |\psi\rangle - \delta\tau(\hat{H} - E)|\psi\rangle, \quad (1.48)$$

where $E = \langle\psi|\hat{H}|\psi\rangle/\langle\psi|\psi\rangle$, while the variational update reads

$$|\psi'_{\text{var}}\rangle = |\psi\rangle + \sum_k \delta\theta_k \partial_k |\psi\rangle. \quad (1.49)$$

Since physical states correspond to rays in Hilbert space, the relevant notion of distance is the Fubini-Study distance:

$$d_{\text{FS}}(\psi, \phi) = \arccos\left(\frac{|\langle\psi|\phi\rangle|}{\sqrt{\langle\psi|\psi\rangle\langle\phi|\phi\rangle}}\right). \quad (1.50)$$

The states considered here differ only infinitesimally. Therefore, it is sufficient to work with the second-order expansion

$$d_{\text{FS}}^2(\psi, \phi) = 1 - \frac{|\langle\psi|\phi\rangle|^2}{\langle\psi|\psi\rangle\langle\phi|\phi\rangle} + \mathcal{O}(\delta^4), \quad (1.51)$$

which defines the local metric structure of projective Hilbert space.

Introducing the residual vector

$$|R\rangle = \sum_k \delta\theta_k \partial_k |\psi\rangle + \delta\tau(\hat{H} - E)|\psi\rangle, \quad (1.52)$$

one finds that the Fubini-Study distance between $|\psi'_{\text{var}}\rangle$ and $|\psi'\rangle$ can be expanded to second order as

$$d_{\text{FS}}^2 = \frac{\langle R|R\rangle}{\langle\psi|\psi\rangle} - \frac{|\langle\psi|R\rangle|^2}{\langle\psi|\psi\rangle^2} + \mathcal{O}(\delta\tau^3). \quad (1.53)$$

After straightforward algebra, this expression reduces to the quadratic form

$$d_{\text{FS}}^2 = \sum_{ij} \delta\theta_i^* S_{ij} \delta\theta_j + 2\delta\tau \text{Re} \left[\sum_i \delta\theta_i^* F_i \right] + \delta\tau^2 \text{Var}(E_{\text{loc}}) + \mathcal{O}(\delta\tau^3), \quad (1.54)$$

where

$$S_{ij} = \langle O_i^* O_j \rangle - \langle O_i^* \rangle \langle O_j \rangle \quad (1.55)$$

and

$$F_i = \langle O_i^* E_{\text{loc}} \rangle - \langle O_i^* \rangle \langle E_{\text{loc}} \rangle. \quad (1.56)$$

The optimal update is obtained by minimising the quadratic form with respect to $\delta\theta$. The stationarity condition

$$\frac{\partial d_{\text{FS}}^2}{\partial \delta\theta_i^*} = 0 \quad (1.57)$$

immediately yields

$$\sum_j S_{ij} \delta\theta_j = -\delta\tau F_i. \quad (1.58)$$

This is the stochastic reconfiguration (SR) equation.

Substituting the optimal displacement back into the quadratic form yields the minimum Fubini-Study distance

$$d_{\text{FS},\min}^2 = \delta\tau^2 [\text{Var}(E_{\text{loc}}) - \mathbf{F}^\dagger \mathbf{S}^{-1} \mathbf{F}] + \mathcal{O}(\delta\tau^3). \quad (1.59)$$

The quantity $\text{Var}(E_{\text{loc}})$ measures the squared norm of the exact imaginary-time vector field generated by $(\hat{H} - E)|\psi\rangle$, while the term $\mathbf{F}^\dagger \mathbf{S}^{-1} \mathbf{F}$ quantifies the component of this evolution that can be represented within the tangent space of the variational manifold. Therefore, the residual distance after optimisation measures the projection error associated with the variational approximation.

Two limiting cases are particularly instructive. First, if the variational state is an exact eigenstate of the Hamiltonian, the zero-variance principle implies

$$\text{Var}(E_{\text{loc}}) = 0. \quad (1.60)$$

In this case,

$$F_i = 0, \quad (1.61)$$

and the imaginary-time vector field vanishes identically. Then,

$$d_{\text{FS}}^2 = 0, \quad (1.62)$$

reflecting the fact that the state is a fixed point of the imaginary-time dynamics. Second, if the exact imaginary-time direction belongs to the tangent space of the variational manifold, the projection becomes exact and one finds

$$\mathbf{F}^\dagger \mathbf{S}^{-1} \mathbf{F} = \text{Var}(E_{\text{loc}}). \quad (1.63)$$

The residual distance thus again vanishes. In this case, stochastic reconfiguration exactly reproduces the imaginary-time evolution despite the restriction to the variational manifold. More generally, the difference

$$\text{Var}(E_{\text{loc}}) - \mathbf{F}^\dagger \mathbf{S}^{-1} \mathbf{F} \quad (1.64)$$

provides a quantitative measure of the inability of the variational manifold to represent the exact imaginary-time flow.

The SR Eq. (1.58) derived above leads to the following discrete update rule:

$$\boldsymbol{\theta}_{n+1} = \boldsymbol{\theta}_n - \delta\tau \mathbf{S}^{-1} \mathbf{F}, \quad (1.65)$$

where, upon noting that $\nabla_{\boldsymbol{\theta}} E_{\boldsymbol{\theta}} = 2\text{Re}[\mathbf{F}]$, it becomes transparent that it amounts to the trivial gradient-descent update of Eq. (1.35) preconditioned by the matrix $\mathbf{S}$.

The SR update may therefore be viewed as a gradient-descent step preconditioned by the matrix $\mathbf{S}$. Unlike ordinary gradient descent, however, the preconditioner is not chosen heuristically but arises directly from the geometry of the variational manifold. Directions

corresponding to large physical changes of the wavefunction are automatically suppressed, while directions producing only small physical changes are amplified.

The geometric origin of this construction becomes apparent by considering the infinitesimal distance between neighbouring variational states,

$$\delta s^2 = \sum_{ij} \delta \theta_i^* S_{ij} \delta \theta_j. \quad (1.66)$$

The matrix S therefore defines the metric tensor induced on the variational manifold by the Hilbert-space geometry. More precisely, it corresponds to the real part of the quantum geometric tensor, or equivalently to the pullback of the Fubini–Study metric onto the variational manifold.

Stochastic reconfiguration may therefore be interpreted as a natural-gradient method in which optimisation is performed with respect to distances between quantum states rather than distances between parameter vectors.

1.V The variational Monte Carlo algorithm

We can now assemble the various ingredients introduced above into a complete variational Monte Carlo algorithm. Given a variational Ansatz $\psi_{\theta}(x)$ and a sampler, the objective is to iteratively improve the parameters so as to minimise the variational energy.

Each optimisation step consists of four main operations:

- (i) Generate a set of configurations using Monte Carlo sampling:

$$\{x^{(i)}\}_{i=1}^{N_s} \sim p_{\theta}(x). \quad (1.67)$$

- (ii) Evaluate the local energy and logarithmic derivatives for all sampled configurations:

$$E_{\text{loc}}(x^{(i)}), \quad O_k(x^{(i)}), \quad (1.68)$$

- (iii) Construct Monte Carlo estimates of the force vector

$$F_i = \langle O_i^* E_{\text{loc}} \rangle - \langle O_i^* \rangle \langle E_{\text{loc}} \rangle \quad (1.69)$$

and quantum geometric tensor

$$S_{ij} = \langle O_i^* O_j \rangle - \langle O_i^* \rangle \langle O_j \rangle. \quad (1.70)$$

- (iv) Update the parameters according to the stochastic reconfiguration equation

$$\theta_{n+1} = \theta_n - \eta \mathbf{S}^{-1} \mathbf{F}. \quad (1.71)$$

The procedure is repeated until convergence of the variational energy and other observables of interest.

1.V.1 Energy and wavefunction accuracy

An important consequence of the variational principle is that the energy provides direct information about the quality of the variational wavefunction.

For gapped Hamiltonians, the variational energy provides direct information about the quality of the variational wavefunction. Indeed, denoting by $\Delta = E_1 - E_0$ the many-body gap and by $\mathcal{I}$ the infidelity with respect to the exact ground state, one can show that

$$\mathcal{I} \leq \frac{E_{\theta} - E_0}{\Delta}. \quad (1.72)$$

Then, the error on the expectation value of a generic observable satisfies

$$\delta O = \mathcal{O} \left(\sqrt{\frac{E_{\theta} - E_0}{\Delta}} \right). \quad (1.73)$$

Thus, sufficiently accurate energies generally imply highly accurate estimates of physical observables, explaining why the variational energy is typically the primary quantity monitored during optimisation and benchmarking. Recently, variance-based diagnostics have also been proposed. In particular, the V -score [49] exploits the energy variance to construct a system-independent measure of wavefunction quality accessible directly within variational Monte Carlo.

Chapter 2

Neural quantum states on discrete Hilbert spaces

2.1 Neural quantum states

The discussion so far has been entirely independent of the choice of variational manifold. Variational Monte Carlo only requires a parametrised wavefunction ψ_{θ} whose amplitudes and derivatives can be evaluated efficiently. The remaining question is therefore how to construct sufficiently expressive wavefunction Ansätze for quantum many-body systems.

Historically, a wide variety of variational parametrisations have been developed, including Jastrow wavefunctions, Slater determinants, tensor-network states, and correlated basis-function approaches. In these lectures, we focus on *neural quantum states* (NQS), in which the wavefunction is represented by a neural network.

Given a configuration x in a computational basis, a neural quantum state defines a map

$$x \mapsto \psi_{\theta}(x), \quad (2.1)$$

where θ denotes the weights and biases of the neural network. The variational manifold is therefore specified by the choice of architecture, while the optimisation machinery introduced in the previous sections remains unchanged. Note how unlike conventional machine-learning scenarios, the target quantity is generally complex-valued

In practice, neural quantum states are almost universally formulated in terms of the logarithm of the wavefunction,

$$\log \psi_{\theta}(x) = f_{\theta}(x), \quad (2.2)$$

with

$$\psi_{\theta}(x) = \exp[f_{\theta}(x)]. \quad (2.3)$$

This choice offers several advantages. First, the logarithmic derivatives entering the gradient and stochastic reconfiguration equations become particularly simple:

$$O_k(x) = \partial_{\theta_k} \log \psi_{\theta}(x) = \partial_{\theta_k} f_{\theta}(x). \quad (2.4)$$

The quantities required for optimisation can be obtained directly from the neural-network output without differentiating an exponential. Second, many physically relevant wavefunctions

exhibit amplitudes spanning several orders of magnitude, making direct manipulation of ψ_θ susceptible to overflow and underflow. Representing $\log \psi_\theta$ instead therefore improves numerical stability and dynamic range. In particular, the wavefunction ratios entering the local energy (1.17) and Metropolis-Hastings acceptance probabilities (1.33) can be evaluated as exponentials of logarithmic differences, avoiding the explicit manipulation of exponentially large or small amplitudes. Finally, the decomposition

$$f_\theta(x) = \log |\psi_\theta(x)| + i\varphi_\theta(x) \quad (2.5)$$

naturally separates amplitude and phase information, allowing each contribution to be modelled independently when desired.

2.1.1 Restricted Boltzmann machines

Restricted Boltzmann machines (RBMs) are a class of generative models popularised in the machine-learning literature by Geoffrey Hinton and collaborators. In their seminal work [15], Carleo and Troyer introduced them as variational wavefunctions for quantum many-body systems, initiating the modern field of neural quantum states. Although largely superseded by more expressive architectures, RBMs remain a useful conceptual starting point.

For simplicity, let us consider a system of N spin-1/2 degrees of freedom and choose the computational basis formed by the simultaneous eigenstates of the Pauli operators $\hat{\sigma}_i^z$. A basis configuration is then specified by

$$\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_N), \quad \sigma_i \in \{-1, +1\}, \quad (2.6)$$

and the neural quantum state represents the corresponding wavefunction amplitudes $\psi_\theta(\boldsymbol{\sigma})$. The RBM augments these physical spins by introducing a set of M auxiliary hidden spins

$$\mathbf{h} = (h_1, \dots, h_M), \quad h_j \in \{-1, +1\}. \quad (2.7)$$

The architecture is bipartite: physical and hidden spins are connected, while spins belonging to the same layer do not interact. The model is specified through the energy function

$$E_{\text{RBM}}(\boldsymbol{\sigma}, \mathbf{h}) = -\sum_i a_i \sigma_i - \sum_j b_j h_j - \sum_{ij} W_{ij} \sigma_i h_j, \quad (2.8)$$

where a_i and b_j denote visible and hidden biases and W_{ij} are variational parameters coupling physical and hidden spins. As we will throughout this lecture, we denote $\boldsymbol{\theta} = (\mathbf{a}, \mathbf{b}, \mathbf{W})$ the complete set of parameters of the Ansatz.

The corresponding wavefunction is obtained by summing over all hidden-spin configurations,

$$\psi_\theta(\boldsymbol{\sigma}) = \sum_{\mathbf{h}} e^{-E_{\text{RBM}}(\boldsymbol{\sigma}, \mathbf{h})}. \quad (2.9)$$

Thanks to the absence of hidden-hidden interactions, this summation factorises and can be performed analytically:

$$\psi_\theta(\boldsymbol{\sigma}) = e^{\sum_i a_i \sigma_i} \prod_j \sum_{h_j = \pm 1} \exp \left[h_j \left(b_j + \sum_i W_{ij} \sigma_i \right) \right] \quad (2.10)$$

$$\propto e^{\sum_i a_i \sigma_i} \prod_j \cosh \left(b_j + \sum_i W_{ij} \sigma_i \right), \quad (2.11)$$

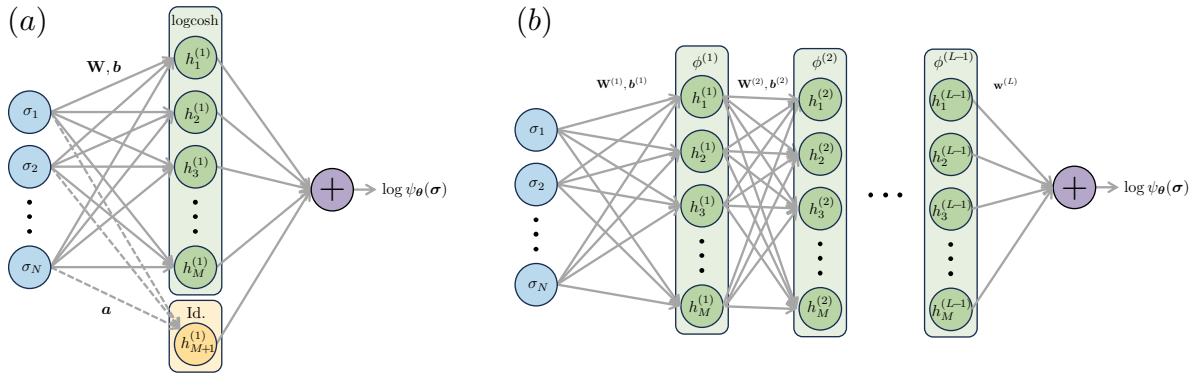


Figure 2.1: (a) Restricted Boltzmann machine (RBM). Analytically marginalising the hidden binary variables reveals the structure of a two-layer perceptron. (b) Deep feed-forward neural network.

where an overall factor 2^M was absorbed into the irrelevant norm of the wavefunction. Taking the logarithm yields

$$\log \psi_{\theta}(\sigma) = \sum_i a_i \sigma_i + \sum_j \log \left[\cosh \left(b_j + \sum_i W_{ij} \sigma_i \right) \right]. \quad (2.12)$$

The hidden spins therefore generate effective many-body correlations between the physical degrees of freedom. Increasing the number of hidden spins systematically enlarges the variational manifold and increases the expressive power of the Ansatz.

This simple analytical form of the wavefunction allows for closed-form expressions for the logarithmic derivatives entering the optimisation procedure:

$$\frac{\partial \log \psi_{\theta}}{\partial a_i} = \sigma_i, \quad \frac{\partial \log \psi_{\theta}}{\partial b_j} = \tanh \left(b_j + \sum_i W_{ij} \sigma_i \right), \quad \frac{\partial \log \psi_{\theta}}{\partial W_{ij}} = \sigma_i \tanh \left(b_j + \sum_k W_{kj} \sigma_k \right). \quad (2.13)$$

This analytical tractability is one of the reasons behind the early success of RBMs as neural quantum states.

2.1.2 Feed-forward neural networks

Eq. (2.12) reveals a close connection between restricted Boltzmann machines and conventional neural networks. Indeed, upon defining

$$h_j^{(1)} = \phi \left(b_j + \sum_i W_{ij} \sigma_i \right), \quad (2.14)$$

with activation function

$$\phi(x) = \log \cosh x, \quad (2.15)$$

one has

$$\log \psi_{\theta}(\sigma) = \sum_i a_i \sigma_i + \sum_j h_j^{(1)}. \quad (2.16)$$

The quantities $h_j^{(1)}$ may therefore be interpreted as the activations of a hidden layer obtained by applying a nonlinear transformation to affine combinations of the input spins. From this perspective, an RBM is simply a particular class of neural network with a single-hidden layer, i.e. a two-layer perceptron. This is schematically illustrated in Fig. 2.1(a).

Modern machine learning has demonstrated that substantially greater representational power can be obtained by composing multiple such layers of nonlinear transformations. This naturally motivates the replacement of RBMs by deeper and more general feed-forward neural networks.

These consist of a sequence of hidden representations

$$\mathbf{h}^{(\ell+1)} = \phi^{(\ell)}(W^{(\ell)}\mathbf{h}^{(\ell)} + \mathbf{b}^{(\ell)}), \quad \mathbf{h}^{(0)} = \boldsymbol{\sigma}, \quad (2.17)$$

where $W^{(\ell)}$, $\mathbf{b}^{(\ell)}$, and $\phi^{(\ell)}$ denote the weights, biases, and activation function of layer ℓ , respectively. After $L-1$ layers, the network produces a scalar output¹

$$\log \psi_{\boldsymbol{\theta}}(\boldsymbol{\sigma}) = \mathbf{w}^{(L)} \cdot \mathbf{h}^{(L-1)}. \quad (2.18)$$

This is schematically illustrated in Fig. 2.1(b).

The expressive power of the network can be increased systematically by increasing its width, corresponding to the number of neurons per layer, or its depth, corresponding to the number of successive nonlinear transformations. This flexibility, together with the availability of efficient automatic-differentiation techniques, has made feed-forward neural networks one of the most widely used building blocks of modern neural quantum states.

2.11 Symmetries and equivariance

Physical wavefunctions often inherit important structural properties from the symmetries of the underlying Hamiltonian. Incorporating these symmetries into a neural quantum state can both restrict the variational search to the physically relevant sector of Hilbert space and improve representational efficiency.

To illustrate these ideas, we consider the one-dimensional transverse-field Ising model

$$\hat{H} = -J \sum_i \hat{\sigma}_i^z \hat{\sigma}_{i+1}^z - h \sum_i \hat{\sigma}_i^x, \quad (2.19)$$

with periodic boundary conditions and N sites. We will work in the σ^z computational basis. This Hamiltonian possesses two important symmetries. First, it is invariant under lattice translations. Second, it is invariant under the global spin-flip transformation $\hat{U} = \prod_i \hat{\sigma}_i^x$, which maps σ_i into $-\sigma_i$, $\forall i$.

An important question is whether these symmetries should be incorporated into the variational Ansatz. Since the exact ground state is symmetric, one might expect the optimisation procedure to discover the symmetry automatically.

To understand why this is not always the case, let us consider the ferromagnetic regime $h \ll J$. The exact finite-size ground state belongs to the even $\mathbb{Z}_2$ sector and is approximately given by

$$|\psi_0\rangle \simeq \frac{|\uparrow\uparrow\cdots\uparrow\rangle + |\downarrow\downarrow\cdots\downarrow\rangle}{\sqrt{2}}. \quad (2.20)$$

¹Note how the last layer typically does not involve any activation function nor bias.

An unconstrained neural quantum state, however, may converge to a wavefunction that predominantly resembles one of the two symmetry-broken components, namely $|\uparrow\uparrow\cdots\uparrow\rangle$ or $|\downarrow\downarrow\cdots\downarrow\rangle$. While having an energy very close to that of the exact finite-size ground state, such a state possesses a finite magnetisation,

$$\left\langle \sum_i \hat{\sigma}_i^z \right\rangle \neq 0. \quad (2.21)$$

This is problematic since the global spin-flip symmetry implies that the expectation value of the total magnetisation must vanish for every finite-size eigenstate of the Hamiltonian². As we have just seen, an unconstrained optimisation may converge to a low-energy state that reproduces the energy accurately while yielding qualitatively incorrect predictions for physically relevant observables. In the present example, one might incorrectly conclude that the system possesses a finite magnetisation.

The problem is further exacerbated by Monte Carlo sampling. When local Metropolis-Hastings updates are employed, transitions between symmetry-related sectors may become exponentially rare, causing the Markov chain to remain trapped within a single sector for long periods of time. As a consequence, symmetry-sensitive observables can exhibit large finite-sampling biases even when the variational state itself is nearly symmetric.

2.II.1 Symmetry projection

A simple and common strategy to circumvent the above issues is to project an arbitrary variational wavefunction onto the desired symmetry sector. The resulting wavefunction automatically satisfies the corresponding symmetry constraints while remaining parametrised by the same underlying neural network.

For our example, given an arbitrary wavefunction ψ_{θ} , we might obtain an even-parity state as the even combination of the amplitudes of the configuration σ and its spin-flipped counterpart $-\sigma$ as

$$\psi_{\theta}^{(+)}(\sigma) = \psi_{\theta}(\sigma) + \psi_{\theta}(-\sigma), \quad (2.22)$$

which correctly belongs to the even $\mathbb{Z}_2$ sector. Indeed, we can readily check that the output is symmetry invariant $\psi_{\theta}^{(+)}(\sigma) = \psi_{\theta}^{(+)}(-\sigma)$. The same idea applies to translation symmetry. Let $\hat{T}$ denote the lattice translation operator, then the wavefunction belonging to the momentum sector k is obtained as

$$\psi_{\theta}^{(k)}(\sigma) = \sum_{n=0}^{N-1} e^{-ikn} \psi_{\theta}(T^n \sigma). \quad (2.23)$$

More generally, given a finite symmetry group G acting on configurations according to $x \mapsto gx$, one may project onto a symmetry sector characterised by the character $\chi(g)$ through [50]

$$\psi_{\theta}^{(\chi)}(x) = \sum_{g \in G} \chi(g)^* \psi_{\theta}(gx). \quad (2.24)$$

This construction is exact and can be applied to essentially any neural-network architecture. Its computational cost, however, scales linearly with the size of the symmetry group, since the

²If $\hat{U}$ denotes the spin-flip operator, one has $[\hat{H}, \hat{U}] = 0$ and $\hat{U}^\dagger \hat{M} \hat{U} = -\hat{M}$, where $\hat{M} = \sum_i \hat{\sigma}_i^z$. Therefore, $\langle \psi_n | \hat{M} | \psi_n \rangle = 0$ for any energy eigenstate $|\psi_n\rangle$.

underlying network must be evaluated for every symmetry-related configuration. Therefore, it is most often used only for global discrete symmetries and point groups.

2.11.2 Equivariant architectures

While the computational overhead of symmetry projection is often manageable for global discrete symmetries, it can become limiting for lattice translations and other large groups. An alternative strategy consists in incorporating the symmetry directly into the architecture, by constructing neural networks whose intermediate representations transform predictably under the action of the symmetry group. Such architectures are said to be *equivariant*.

More precisely, a map f is equivariant under a group G if

$$f(gx) = \rho(g)f(x), \quad (2.25)$$

for all $g \in G$, where $\rho(g)$ denotes the corresponding action of the group on the feature space. Crucially, equivariance is preserved under composition. Indeed, if

$$f(gx) = \rho(g)f(x), \quad h(gz) = \eta(g)h(z), \quad (2.26)$$

then

$$(h \circ f)(gx) = \eta(g)(h \circ f)(x), \quad (2.27)$$

so that the composition remains equivariant. Hence, one may construct deep equivariant neural networks by simply stacking equivariant layers.

A simple example is provided by lattice translations. Consider a spin configuration σ of our one-dimensional Ising system. A convolutional layer computes a collection of feature maps $h_{i\alpha}^{(\ell)}$, where i labels the lattice site and $\alpha = 1, \dots, C_\ell$ labels the feature channel, with C_ℓ the number of channels at layer ℓ . Starting from $h_{i1}^{(0)} = \sigma_i$, successive feature maps are constructed according to

$$h_{i\alpha}^{(\ell+1)} = \phi \left(\sum_{\beta=1}^{C_\ell} \sum_{r=-R}^R W_{\alpha\beta r}^{(\ell)} h_{i+r,\beta}^{(\ell)} + b_\alpha^{(\ell)} \right), \quad (2.28)$$

where $W_{\alpha\beta r}^{(\ell)}$ denotes the convolution kernel connecting input channel β to output channel α , $b_\alpha^{(\ell)}$ is a bias term, R is the convolutional radius determining the range of the filter, and ϕ is a nonlinear activation function. Since the same kernels are applied at every lattice site, translating the input configuration simply translates the resulting feature maps,

$$h^{(\ell)}(T\sigma) = Th^{(\ell)}(\sigma), \quad (2.29)$$

and the layer is therefore equivariant under lattice translations.

By composing multiple convolutional layers, one obtains a hierarchy of translation-equivariant feature maps. A translation-invariant wavefunction can then be constructed through a final pooling operation,

$$\log \psi_{\theta}^{k=0}(\sigma) = \sum_{\alpha=1}^{C_L} w_\alpha \left(\frac{1}{N} \sum_{i=1}^N h_{i\alpha}^{(L)} \right), \quad (2.30)$$

where w_α are the additional variational parameters of the visible layer that projects the invariant feature vector into a scalar log amplitude³.

The key advantage over explicit symmetry projection is that the network is evaluated only once.

³Note how projecting onto an arbitrary momentum sector now becomes straightforward by replacing $\sum_{i=n}^N h_{n\alpha}^{(L)} \mapsto \sum_{i=n}^N e^{-ikn} h_{n\alpha}^{(L)}$.

Chapter 3

Neural quantum states for particle systems

3.1 Particle systems

Up to now, we have focused primarily on lattice systems described by discrete spin configurations $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_N)$. Many quantum systems of practical interest, however, are more naturally formulated in terms of particles moving in continuous space. Examples include ultracold atomic gases, liquid helium, and electronic systems. In this setting, the many-body wavefunction becomes a function of the particle coordinates, $\mathbf{R} = (\mathbf{r}_1, \dots, \mathbf{r}_N)$, where $\mathbf{r}_i \in \mathbb{R}^d$ denotes the position of the i th particle.

Compared to lattice systems, this setting introduces two new challenges. First, the wavefunction must now be represented as a function of continuous particle coordinates. Second, identical particles are fundamentally indistinguishable, implying that physical wavefunctions must satisfy additional constraints imposed by quantum statistics. In this section, we discuss how neural quantum states can address both of these challenges through the example of bosonic systems.

3.1.1 Bosonic wavefunctions

The defining property of bosons is that their wavefunction remains invariant under the exchange of any pair of particles,

$$\psi(\mathbf{r}_{\pi(1)}, \dots, \mathbf{r}_{\pi(N)}) = \psi(\mathbf{r}_1, \dots, \mathbf{r}_N), \quad (3.1)$$

for every permutation π .

In contrast to the lattice symmetries discussed previously, this constraint is not merely a useful inductive bias but a fundamental physical requirement. An unconstrained neural network can represent arbitrary functions of the particle coordinates, including functions that violate bosonic exchange symmetry. Such wavefunctions do not belong to the physical Hilbert space and may therefore exhibit variational energies lower than the exact bosonic ground-state energy. Therefore, permutation symmetry must be enforced explicitly at the level of the variational Ansatz.

At first sight, one might attempt to achieve this through the symmetry-projection techniques introduced in the previous section. Given an arbitrary wavefunction $\psi_{\boldsymbol{\theta}}$, a bosonic wavefunction can indeed be obtained through the symmetrisation

$$\psi_{\boldsymbol{\theta}}^{(\text{B})}(\mathbf{R}) = \sum_{\pi \in S_N} \psi_{\boldsymbol{\theta}}(\pi \mathbf{R}), \quad (3.2)$$

where S_N denotes the permutation group of N particles. Unlike the translation or point-group symmetries encountered previously, however, the permutation group contains $|S_N| = N!$ elements. The computational cost of explicit symmetrisation therefore grows factorially with the number of particles and rapidly becomes prohibitive.

For this reason, modern neural quantum states typically incorporate permutation symmetry directly into the architecture by constructing permutation-invariant representations of the particle coordinates.

3.1.2 Deep sets

The simplest strategy to construct a bosonic neural quantum state consists in building permutation invariance directly into the architecture. Under suitable assumptions, any continuous permutation-invariant function can be decomposed as [51, 52]

$$f(\mathbf{R}) = \rho\left(\sum_{i=1}^N \phi(\mathbf{r}_i)\right), \quad (3.3)$$

for appropriate functions ϕ and ρ . Upon replacing these functions by a variational approximation, this becomes the so-called Deep Sets architecture.

One first computes a feature vector for each particle,

$$\mathbf{h}_i = \phi(\mathbf{r}_i), \quad (3.4)$$

using a shared neural network. Since the same network is applied to every particle, the resulting features transform equivariantly under permutations. A permutation-invariant representation is then obtained through a pooling operation,

$$\mathbf{h} = \sum_{i=1}^N \mathbf{h}_i, \quad (3.5)$$

after which a second neural network produces the wavefunction amplitude:

$$\log \psi_{\theta}(\mathbf{R}) = \rho(\mathbf{h}). \quad (3.6)$$

Because the pooling operation is invariant under particle permutations, the resulting wavefunction automatically satisfies the bosonic exchange symmetry.

Deep sets provide a simple and computationally efficient route towards permutation-invariant neural quantum states [20, 21]. Their main limitation, however, is that particles are processed independently before the final pooling step, limiting the architecture's ability to represent the strong inter-particle correlations that characterise interacting quantum systems.

3.1.3 Message-passing neural networks

A natural improvement consists in allowing particles to exchange information before the final aggregation step. This idea forms the basis of *message-passing neural networks* [25].

The system is represented as a graph whose nodes correspond to particles and whose edges encode pairwise geometric information. Each particle is associated with a feature vector $\mathbf{h}_i^{(\ell)}$,

which is iteratively updated according to information received from the other particles. A generic message-passing layer takes the form

$$\mathbf{m}_i^{(\ell)} = \sum_{j \neq i} M_{\boldsymbol{\theta}}^{(\ell)}(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{r}_{ij}), \quad (3.7)$$

$$\mathbf{h}_i^{(\ell+1)} = U_{\boldsymbol{\theta}}^{(\ell)}(\mathbf{h}_i^{(\ell)}, \mathbf{m}_i^{(\ell)}), \quad (3.8)$$

where

$$\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j \quad (3.9)$$

denotes the relative coordinate between particles i and j , while $M_{\boldsymbol{\theta}}^{(\ell)}$ and $U_{\boldsymbol{\theta}}^{(\ell)}$ are learnable message and update functions. Because the messages depend only on relative coordinates, the resulting architecture is naturally invariant under global translations.

The summation over particles is permutation invariant, implying that the particle features transform equivariantly under particle exchanges. Therefore, permutation symmetry is preserved throughout the network. At the same time, the iterative exchange of information allows the architecture to build increasingly complex many-body correlations.

After a number of message-passing steps, a permutation-invariant wavefunction can be constructed through a final pooling operation,

$$\log \psi_{\boldsymbol{\theta}}(\mathbf{R}) = \rho_{\boldsymbol{\theta}}\left(\sum_i \mathbf{h}_i^{(L)}\right), \quad (3.10)$$

where $\rho_{\boldsymbol{\theta}}$ is a learnable function. Message-passing architectures therefore combine the exact symmetry requirements of bosonic wavefunctions with a considerably richer representation of inter-particle correlations than Deep Sets. They form the basis of many modern neural quantum states for continuous-space systems.

3.1.4 Physical priors

While neural networks provide highly expressive variational Ansätze, learning all physically relevant correlations from scratch can be inefficient. In continuous-space systems, significant improvements can often be obtained by integrating known physical properties of the wavefunction directly into the parametrisation.

A particularly common construction consists in factorising the wavefunction as

$$\Psi_{\boldsymbol{\theta}}(\mathbf{R}) = e^{U(\mathbf{R})} \Phi_{\boldsymbol{\theta}}(\mathbf{R}), \quad (3.11)$$

where $\Phi_{\boldsymbol{\theta}}$ is a neural-network wavefunction and

$$U(\mathbf{R}) = \sum_{i < j} u(r_{ij}) \quad (3.12)$$

is a pairwise Jastrow factor. The function $u(r)$ is chosen to reproduce known short-range correlations between particles. By explicitly incorporating part of the physics into the Ansatz, the neural network can focus on learning the remaining many-body correlations, often resulting in improved optimisation and reduced statistical variance.

An important application of this idea arises in systems with singular interactions. Consider two particles interacting through a potential $V(r)$ that diverges as $r \rightarrow 0$. Since the local energy contains both kinetic and potential contributions,

$$E_{\text{loc}}(\mathbf{R}) = -\frac{1}{2\mu} \frac{\nabla^2 \psi(\mathbf{R})}{\psi(\mathbf{R})} + V(\mathbf{R}), \quad (3.13)$$

where $\mu = 1/2$ is the reduced mass¹, the wavefunction must develop a specific short-distance behaviour such that the divergence of the kinetic energy cancels that of the potential. This requirement is known as a *cusp condition*.

For example, for the Coulomb interaction

$$V(r) = \frac{q_1 q_2}{r}, \quad (3.14)$$

the exact wavefunction satisfies the so-called Kato cusp condition [53]

$$\left. \frac{\partial_r \Psi(r)}{\Psi(r)} \right|_{r=0} = \mu q_1 q_2, \quad (3.15)$$

where μ denotes the reduced mass. This implies the short-distance behaviour

$$\Psi(r) \propto e^{\mu q_1 q_2 r}, \quad (3.16)$$

which can be enforced directly through the Jastrow factor. More generally, cusp conditions provide a systematic way of exploiting known asymptotic properties of the wavefunction into neural quantum states, improving both their physical fidelity and optimisation stability.

3.II Fermionic systems

For bosonic systems, the main challenge was to construct wavefunctions satisfying the appropriate exchange symmetries. Fermionic systems introduce a more fundamental difficulty: the wavefunction must be antisymmetric under particle exchange. This antisymmetry gives rise to a non-trivial sign structure that lies at the heart of the fermionic many-body problem.

3.II.1 Fermionic Hilbert spaces

Consider a system of spinless fermions on a lattice of L sites. Because of the Pauli exclusion principle, each site can be either empty or occupied,

$$n_i \in \{0, 1\}. \quad (3.17)$$

A many-body basis state is therefore specified by an occupation pattern

$$\mathbf{n} = (n_1, \dots, n_L). \quad (3.18)$$

¹We here use unit masses as a convention $m_i = 1$, which yields $\mu = m_1 m_2 / (m_1 + m_2) = 1/2$.

At first sight, this *Fock* basis appears identical to that of a spin-1/2 system. The crucial difference is that fermionic states are constructed from creation operators satisfying the canonical anticommutation relations

$$\{\hat{c}_i, \hat{c}_j^\dagger\} = \delta_{ij}, \quad \{\hat{c}_i, \hat{c}_j\} = \{\hat{c}_i^\dagger, \hat{c}_j^\dagger\} = 0. \quad (3.19)$$

In particular,

$$\hat{c}_i^\dagger \hat{c}_j^\dagger = -\hat{c}_j^\dagger \hat{c}_i^\dagger, \quad (3.20)$$

so exchanging two fermions changes the sign of the many-body state. As we shall see, this simple property is responsible for much of the complexity of fermionic quantum many-body systems.

3.II.2 Canonical ordering

Because fermionic creation operators anticommute, the order in which they are applied matters. For example,

$$\hat{c}_1^\dagger \hat{c}_3^\dagger |\mathbf{n}\rangle = -\hat{c}_3^\dagger \hat{c}_1^\dagger |\mathbf{n}\rangle. \quad (3.21)$$

Although both expressions correspond to particles occupying sites 1 and 3, they differ by a sign.

To remove this ambiguity, basis states are defined using a fixed ordering convention,

$$|\mathbf{n}\rangle = (\hat{c}_1^\dagger)^{n_1} (\hat{c}_2^\dagger)^{n_2} \dots (\hat{c}_L^\dagger)^{n_L} |0\rangle, \quad (3.22)$$

where $|0\rangle$ denotes the vacuum state, satisfying $\hat{c}_i |0\rangle = 0$, $\forall i$. This choice is known as the *canonical ordering*. Once the ordering convention has been fixed, every occupation pattern uniquely defines a many-body basis state. The occupation-number basis therefore provides a convenient representation of fermionic wavefunctions,

$$|\psi\rangle = \sum_{\mathbf{n}} \psi(\mathbf{n}) |\mathbf{n}\rangle, \quad (3.23)$$

where the amplitudes $\psi(\mathbf{n})$ are defined with respect to the canonically ordered basis.

Although the basis states are labelled by binary occupation numbers, the fermionic exchange statistics have not disappeared. Rather, they are encoded implicitly in the action of the creation and annihilation operators on the canonically ordered basis. This hidden sign structure distinguishes fermions from spin systems and plays a central role in the construction of fermionic neural quantum states.

Maintaining a consistent ordering is therefore crucial in simulations. When an operator acts on a basis state, creation and annihilation operators must generally be rearranged into canonical order. Each exchange produces a minus sign, so matrix elements acquire a phase determined by the parity of the corresponding permutation. Consider for example the hopping operator $\hat{c}_3^\dagger \hat{c}_1$ acting onto a Fock state,

$$\begin{aligned} \hat{c}_3^\dagger \hat{c}_1 |1,1,0,0\rangle &= \hat{c}_3^\dagger \hat{c}_1 \hat{c}_1^\dagger \hat{c}_2^\dagger |0\rangle \\ &= \hat{c}_3^\dagger \hat{c}_2^\dagger |0\rangle \\ &= -\hat{c}_2^\dagger \hat{c}_3^\dagger |0\rangle \\ &= -|0,1,1,0\rangle. \end{aligned} \quad (3.24)$$

The minus sign appears when restoring the creation operators to canonical order. This is the elementary origin of the fermionic sign structure.

The amplitudes of a fermionic wavefunction must therefore reproduce a highly non-trivial pattern of positive and negative signs. Constructing variational wavefunctions with the correct sign structure is one of the central challenges of fermionic quantum many-body simulation.

3.11.3 Slater determinants

A remarkably simple solution to this problem is provided by Slater determinants. Rather than reproducing the fermionic signs configuration by configuration, the antisymmetry is enforced exactly at the level of the wavefunction Ansatz.

Consider a system of non-interacting fermions described by

$$\hat{H} = \sum_{ij} h_{ij} \hat{c}_i^\dagger \hat{c}_j. \quad (3.25)$$

Diagonalising the single-particle Hamiltonian h_{ij} yields eigenenergies ϵ_α and associated orbitals $\phi_\alpha(i)$ with the corresponding creation operators

$$\hat{d}_\alpha^\dagger = \sum_i \phi_\alpha(i) \hat{c}_i^\dagger. \quad (3.26)$$

The many-body ground state is obtained by occupying the N lowest-energy orbitals,

$$|\psi_{\text{SD}}\rangle = \hat{d}_1^\dagger \hat{d}_2^\dagger \cdots \hat{d}_N^\dagger |0\rangle. \quad (3.27)$$

Expanding this state in the occupation-number basis yields

$$\begin{aligned} \psi_{\text{SD}}(i_1, \dots, i_N) &= \det[\phi_\alpha(i_\beta)] \\ &= \det \begin{pmatrix} \phi_1(i_1) & \cdots & \phi_1(i_N) \\ \vdots & \ddots & \vdots \\ \phi_N(i_1) & \cdots & \phi_N(i_N) \end{pmatrix}, \end{aligned} \quad (3.28)$$

known as a *Slater determinant*.

Exchanging two particle coordinates corresponds to exchanging two columns of the determinant and therefore changes its sign, thereby reproducing the fermionic sign structure exactly. For this reason, Slater determinants form the foundation of most fermionic variational Ansätze.

For variational Monte Carlo, however, wavefunctions are evaluated in the occupation-number basis. Thanks to the canonical ordering, every configuration $\mathbf{n}$ containing N fermions can be identified with the ordered set of occupied sites

$$\mathbf{n} \longleftrightarrow (i_1, \dots, i_N), \quad i_1 < \cdots < i_N. \quad (3.29)$$

The corresponding Slater amplitude is obtained by selecting the columns of the orbital matrix associated with the occupied sites and evaluating the resulting determinant.

3.II.4 Correlated fermionic wavefunctions

Slater determinants provide an exact description of non-interacting fermions but neglect interaction-induced correlations. A natural improvement consists in augmenting the determinant with a permutation symmetric correlation factor,

$$\psi_{\phi,\theta}(\mathbf{n}) = e^{J_\theta(\mathbf{n})} \psi_{\text{SD},\phi}(\mathbf{n}), \quad (3.30)$$

where $J_\theta(\mathbf{n})$ is known as a Jastrow factor. Because the Jastrow factor is symmetric under particle exchange, it does not modify the fermionic sign structure. The determinant therefore remains entirely responsible for the signs of the wavefunction, while the Jastrow factor captures correlations. While the Jastrow factor typically takes the form of a two-body potential, $J_\theta(\mathbf{n}) = \sum_{i<j} n_i W_{ij} n_j$, neural networks can be incorporated naturally within this framework by replacing it with a neural-network Ansatz.

Such neural Jastrow-Slater wavefunctions often provide a substantial improvement over simple Slater determinants. However, since the prefactor is strictly positive, it cannot modify the set of configurations,

$$\{\mathbf{n} \mid \psi(\mathbf{n}) = 0\}, \quad (3.31)$$

known as the *nodal structure* of the wavefunction. This set acts as a boundary between regions of the support of the wavefunction with different amplitude signs, which thus far remains entirely determined by the Slater determinant. Improving the nodal structure therefore requires modifying the determinant itself. This idea underlies backflow wavefunctions.

3.II.5 Neural backflow

Rather than multiplying the determinant by a correlation factor, backflow introduces many-body configuration-dependent [54–56],

$$\phi_\alpha(i) \longrightarrow \phi_{\alpha,\theta}(i;\mathbf{n}), \quad (3.32)$$

where $\phi_{\alpha,\theta}$ is parametrised as a neural network.

The resulting wavefunction takes the form

$$\psi_{\text{BF},\theta}(\mathbf{n}) = \det[\phi_{\alpha,\theta}(i_\beta;\mathbf{n})]. \quad (3.33)$$

Since the orbitals now depend on the full configuration, the determinant itself changes from one configuration to another. Backflow transformations can therefore modify both the amplitudes and the nodal structure of the wavefunction. Modern fermionic neural quantum states often implement the backflow transformation using neural networks, yielding highly expressive variational Ansätze capable of describing strongly correlated fermionic systems.

Different physical systems may require different choices of antisymmetric reference state. While Slater determinants provide a natural starting point for weakly interacting fermions, systems exhibiting pairing are often described more efficiently by Bardeen-Cooper-Schrieffer (BCS) wavefunctions and their Pfaffian generalisations. Backflow transformations and neural-network parametrisations can be applied on top of any of these constructions, yielding highly expressive variational Ansätze tailored to the expected phenomenology of the system.

Chapter 4

A modern perspective on NQS design

Throughout these notes we have encountered a variety of neural-quantum-state architecture, ranging from restricted Boltzmann machines and convolutional neural networks to message-passing architectures and neural backflow wavefunctions. Although these Ansätze appear rather different, they are ultimately designed to address the same challenge, namely constructing expressive variational wavefunctions while respecting the physical structure of the underlying quantum problem.

In recent years, the design of neural quantum states has increasingly followed the developments of broader machine learning. Rather than constructing the wavefunction directly from the physical degrees of freedom, modern architectures first learn an effective representation of the configuration before converting this representation into a wavefunction. This separation between representation learning and wavefunction design has emerged as one of the central organising principles of modern neural quantum states.

4.1 Representation learning

Traditional variational Ansätze often operate directly on the configuration of the physical degrees of freedom. For example, a Jastrow factor is constructed from particle coordinates or occupation numbers, a Slater determinant from single-particle orbitals, and a restricted Boltzmann machine directly from the physical configuration.

Modern architectures instead introduce an intermediate feature space. Given a configuration $\mathbf{x}$, a deep neural network first constructs a set of learned representations,

$$\mathbf{x} \mapsto \mathbf{z}_\phi(\mathbf{x}), \quad (4.1)$$

where $\mathbf{z}_\phi$ denotes a collection of learned features parametrised by ϕ . The wavefunction is then obtained from these representations through a second function,

$$\log \psi_\theta(\mathbf{x}) = f_\eta(\mathbf{z}_\phi(\mathbf{x})), \quad (4.2)$$

with variational parameters

$$\theta = (\phi, \eta). \quad (4.3)$$

The central idea is that the feature space may possess a much simpler structure than the original Hilbert space [22, 57]. The role of the deep encoder is therefore to learn representations in which the final wavefunction becomes easier to model.

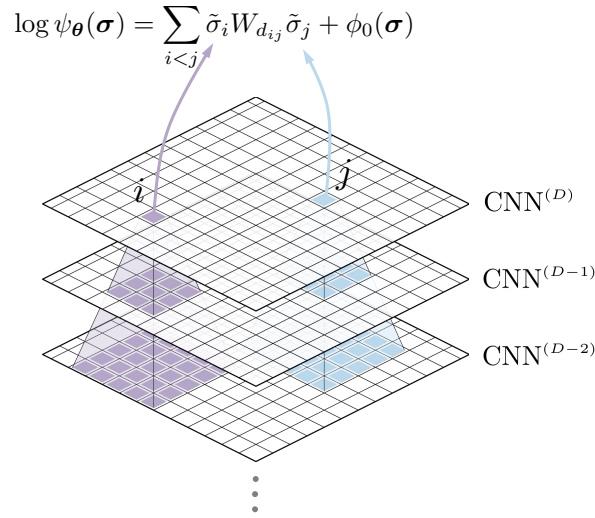


Figure 4.1: Schematic representation of the neural dressed Jastrow architecture of Eq. (4.6). The backflow transformation dresses the bare local quantum numbers entering the Jastrow term with equivariant many-body features. These features bear a dependence on the configuration of all sites within the network’s receptive field. The latter is represented by the shaded region for an exemplary deep convolutional neural network with 3×3 square filters

From this perspective, many modern neural quantum states can be viewed as the combination of two distinct components. The first is a deep encoder responsible for learning useful many-body representations. The second is a physically motivated readout layer that converts these representations into a wavefunction satisfying the desired physical constraints.

4.11 The simplest example: dressed Jastrow wavefunctions

The representation-learning viewpoint is perhaps most easily understood through the example of dressed Jastrow wavefunctions [22]. This Ansatz provides one of the simplest settings in which learned representations can be incorporated into a many-body wavefunction while retaining a clear physical interpretation.

Let us consider a translational invariant system of spins $1/2$ in some basis. Starting from a conventional Jastrow form,

$$|\psi_{\theta}\rangle = e^{\sum_{i<j} W_{d_{ij}} \hat{\sigma}_i^z \hat{\sigma}_j^z} |\phi_0\rangle \quad \Leftrightarrow \quad \log \psi(\boldsymbol{\sigma}) = \sum_{i<j} W_{d_{ij}} \sigma_i \sigma_j + \log \phi_0(\boldsymbol{\sigma}), \quad (4.4)$$

where d_{ij} denotes the distance between sites i and j , and ϕ_0 is some prior we want to improve over, say the mean-field solution. One then replaces the bare spin projections by *dressed* spins,

$$\tilde{\sigma}_i = \sigma_i + \sum_{\mu} a_{\mu} h_{i,\mu}(\boldsymbol{\sigma}). \quad (4.5)$$

Here $h_{i,\mu}$ are local, translation-equivariant features produced by a deep neural network. The resulting wavefunction takes the form

$$\log \psi_{\theta}(\sigma) = \sum_{i < j} W_{d_{ij}} \tilde{\sigma}_i \tilde{\sigma}_j + \log \phi_0(\sigma). \quad (4.6)$$

This construction separates two types of correlations. The Jastrow kernel $W_{d_{ij}}$ captures long-range density correlations in a simple, physically interpretable way, while the neural network dresses each occupation using local many-body information within its receptive field (see Fig. 4.1).

For gapped systems, correlations are expected to have a finite characteristic length scale. The local dressing network can therefore remain essentially size-independent, while the Jastrow term accounts for the residual long-range structure. In a translation-invariant system, the number of independent Jastrow parameters is controlled by the possible displacement vectors d_{ij} and therefore grows with the linear system size, rather than with the total number of sites.

This example illustrates the general principle of modern neural quantum states: a deep encoder learns local effective degrees of freedom, while a simple physically motivated readout captures the dominant structure of the wavefunction and provides the main inductive bias. The complexity of the wavefunction is therefore shifted from the final Ansatz itself to the representation-learning stage.

4.III Vision transformers

A particularly successful example of this paradigm is provided by transformer-based neural quantum states inspired by the Vision Transformer (ViT) architecture [57, 58].

In architectures such as convolutional networks or message-passing neural networks, information propagates locally through the system. Long-range correlations are therefore built progressively through the depth of the network. While this locality is often appropriate for gapped systems, it may become limiting in critical systems, gapless phases, or systems with long-range interactions. In such scenarios, it is interesting to allow distant degrees of freedom to communicate directly.

Transformers address this challenge through *self-attention*. The physical degrees of freedom are first equivariantly mapped to learned embeddings through some parametrised function

$$\mathbf{e}_i = g_{\theta}(\mathbf{x}'_i), \quad (4.7)$$

where g_{θ} is some parametrised map, and $\mathbf{x}'_i$ corresponds to the original i th degree of freedom x_i , optionally augmented with additional equivariant features, for instance for a lattice model one typically feeds a finite patch of configurations around i . These embeddings serve as the initial representation $\mathbf{z}_i^{(0)} = \mathbf{e}_i$.

The network then constructs a hierarchy of increasingly refined representations

$$\mathbf{z}_i^{(0)} \longrightarrow \mathbf{z}_i^{(1)} \longrightarrow \dots \longrightarrow \mathbf{z}_i^{(L)}, \quad (4.8)$$

through a sequence of transformer layers. The key ingredient of each layer is the self-attention mechanism. First, a set of learned queries, keys, and values are produced,

$$\mathbf{z}_i^{(\ell)} \longmapsto \mathbf{q}_i^{(\ell)}, \mathbf{k}_i^{(\ell)}, \mathbf{v}_i^{(\ell)}. \quad (4.9)$$

From these, the representation of every degree of freedom is updated by aggregating information from all others,

$$\mathbf{z}_i^{(\ell+1)} = \sum_j \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)}, \quad (4.10)$$

where the attention coefficients are given by

$$\alpha_{ij}^{(\ell)} = \text{softmax}(\mathbf{q}_i^{(\ell)} \cdot \mathbf{k}_j^{(\ell)}) \quad (4.11)$$

Unlike a fixed interaction graph, the coefficients $\alpha_{ij}^{(\ell)}$ depend on the configuration itself. Self-attention can therefore be interpreted as a learned, configuration-dependent interaction pattern that allows information to be exchanged directly between arbitrary degrees of freedom.

A final readout layer then converts this representation into a wavefunction:

$$\log \psi_{\theta}(\mathbf{x}) = f_{\eta}(\mathbf{z}_i^{(L)}). \quad (4.12)$$

The readout is typically very simple. It may be a good old RBM, Jastrow-like, determinant-based, or a backflow construction. The transformer therefore acts primarily as a representation-learning engine rather than as the wavefunction itself. The role of the transformer is therefore not to replace the wavefunction Ansatz itself, but rather to provide a powerful representation-learning mechanism. As in the neural backflow-Jastrow wavefunction discussed above, the encoder learns useful many-body features while the final readout layer imposes the desired physical structure of the wavefunction.

As in the dressed Jastrow construction discussed previously, these representations are not themselves the wavefunction. Their role is to provide an effective feature space from which the wavefunction can be constructed through a final readout,

$$\log \psi_{\theta}(\mathbf{x}) = f_{\eta}(\mathbf{z}_i^{(L)}). \quad (4.13)$$

The readout may be Jastrow-like, RBM-based, determinant-based, or a neural backflow construction. The transformer therefore acts primarily as a representation-learning engine, while physically motivated readouts encode the desired symmetries, constraints, and inductive biases.

4.IV Outlook

Architectures with strong physical inductive biases, such as convolutional and message-passing networks, often achieve excellent performance when their assumptions match the underlying physics. Transformers offer a complementary approach, providing greater flexibility and direct access to long-range correlations at the cost of weaker prior structure and higher computational complexity.

As large-scale simulations become increasingly accessible, transformer-based neural quantum states have emerged as some of the most accurate and scalable variational Ansätze available, driving many current state-of-the-art results, particularly in systems with strong long-range correlations.

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